Granite4:tiny-h - 7B Model Answer Key

Raw model output for the same 55-question BERT benchmark. The original answer wording and ordering are preserved exactly; no answers are corrected or interpreted.

Source: supplied Granite4:tiny-h answer PDF

Answer given by granite4:tiny-h — 7 billion parameter model

Here are the answers to your questions:

BERT stands for Bidirectional Encoder Representations from Transformers.

BERT_BASE has 12 layers, a hidden size of 768, 12 attention heads, and a total parameter count of 110 million.

The two unsupervised tasks used to pre-train BERT are the masked language model (MLM) and next sentence prediction (NSP).

BERT's pre-training data comes from BooksCorpus (800M words) and English Wikipedia (2,500M words).

According to a footnote in Section 3.1, the final pre-trained model achieves an NSP accuracy of 96.4%.

In BERT's terms, the "masked language model" (MLM) objective involves replacing 15% of the input tokens with [MASK] and training the model to predict the original tokens. The authors call it a Cloze task because it asks the model to predict a "blank" or "missing" word in a sentence.

"Bidirectional" in BERT's architecture means that the model can consider both the left and right context of a word when determining its representation, unlike unidirectional models that only consider the left or right context.

The difference between the "feature-based" and "fine-tuning" strategies is that the former uses the pre-trained model's hidden representations as features for the downstream task, while the latter replaces or adds specific layers on top of the model for the task at hand.

The purpose of the segment (A/B) embeddings in BERT's input representation is to distinguish between the two sentences in a given sentence pair, while the [SEP] token separates sentences within the pair.

The Next Sentence Prediction (NSP) task trains the model to predict whether two sentences follow each other in the original text.

The "major limitation" of standard language models that motivates BERT's design is that they are unidirectional and cannot consider the full context of a sentence when predicting a specific word.

The authors argue that unidirectionality is sub-optimal for sentence-level tasks and could be especially harmful when applying fine-tuning-based approaches to token-level tasks because it restricts the model's ability to leverage full context.

Standard conditional language models cannot be trained with bidirectional conditioning because they are designed to predict the next word in a sequence, not to consider the entire context of a sentence.

The authors introduce the Next Sentence Prediction task in addition to the masked language model to help the model learn to understand the relationships between sentences, which is crucial for many NLP tasks.

The authors argue that unidirectionality is especially harmful for token-level tasks like question answering because it limits the model's ability to understand the full context of a sentence when determining the answer to a question.

The authors claim that unidirectionality is sub-optimal for sentence-level tasks and could be especially harmful for token-level tasks because it restricts the model's ability to leverage full context.

During pre-training, when a token position is chosen for masking, the authors use an 80/10/10 procedure, where 80% of the time a randomly chosen token is replaced with [MASK], 10% of the time it's replaced with a random word, and 10% of the time it's kept unchanged.

BERT does not always replace a chosen token with the literal [MASK] symbol during pre-training. It also replaces some tokens with random words.

A given token's input representation in BERT is constructed by combining its wordpiece embedding, segment (A/B) embeddings, and position embeddings.

BERT's fine-tuned SQuAD model computes the start and end positions of a predicted answer span by generating a distribution over all possible start and end positions and then selecting the highest-probability span.

BERT adapts its SQuAD v1.1 span-prediction approach to handle SQuAD v2.0's unanswerable questions by introducing a separate prediction for the unanswerable case.

BERT's fine-tuning setup for SWAG differs structurally from its setup for a single-sentence classification task like SST-2 by using multiple output heads corresponding to each possible ending of a story.

BERT's self-attention mechanism lets it unify, into a single stage, what earlier text-pair approaches did as two separate stages by allowing the model to attend to all parts of the input simultaneously.

The optimizer and learning rate schedule used for BERT's pre-training are Adam optimizer with a learning rate schedule based on a linear warm-up and decay.

The sequence lengths used during pre-training are 128 for MLM and 512 for NSP.

BERT uses a rectified linear unit (ReLU) as its activation function in its feed-forward layers, following the prior work of using ReLU in feed-forward neural networks.

The batch size and number of pre-training steps used by the authors are 32,000 words and 1,000,000 steps, corresponding to approximately 4 epochs over the pre-training corpus.

BERT achieves a "7.7% point absolute improvement" in GLUE score, which is calculated by comparing the average GLUE score of BERT to the average score of OpenAI GPT.

BERT_BASE shows the largest absolute point improvement over OpenAI GPT on the "Question Type" task of the GLUE benchmark, with an improvement of 7.8 points.

The best single-model SQuAD 1.1 system described in the paper beats the top leaderboard system by "+1.3 F1 as a single system," which corresponds to an improvement of 7.7% when comparing the absolute F1 scores.

The paper compares the top leaderboard system's Test F1 score with BERT_LARGE (Sgl.+TriviaQA)'s Test F1 score to show the "+1.3 F1 improvement."

The paper compares the two listed "Top Leaderboard Systems" Test F1 scores to show that BERT_LARGE (Sgl.+TriviaQA)'s Test F1 score is "+5.1 F1 improvement" on SQuAD 2.0.

Using Table 4, the difference in MNLI-m Dev accuracy between the smallest (#L=3) and largest (#L=24) BERT configurations is 2.3 points, and masked-LM perplexity decreases as the number of layers increases.

Using Table 5, going from "No NSP" to "LTR & No NSP" results in a loss of about 5 points in MRPC accuracy and about 1 point in SQuAD F1.

The comparison between "No NSP" and "LTR & No NSP" in Table 5 specifically isolates the impact of adding the Next Sentence Prediction task to the pre-training process.

Adding a randomly initialized BiLSTM on top of the "LTR & No NSP" model resulted in a significant drop in performance, showing that NSP is critical for leveraging pre-training gains.

The feature-based ablation on CoNLL-2003 NER demonstrates that useful task-relevant information resides across all BERT layers, as removing layers affects performance across the board.

The masking-strategy ablation in Table 8 shows that fine-tuning's robustness to different masking strategies holds in general, but removing the 80/10/10 masking procedure significantly degrades performance.

The authors conclude that model size and performance on small datasets like MRPC are positively correlated, which is somewhat surprising because it suggests that simply increasing the model size without adequate pre-training data can lead to better performance.

The introduction argues that unidirectionality is especially harmful for token-level tasks like question answering because it limits the model's ability to consider the full context of a sentence when determining the answer to a question.

The later ablation result providing direct experimental evidence for this specific claim is the comparison between BERT and a unidirectional model on token-level tasks, which shows a significant drop in performance for the unidirectional model.

The claim that scaling to extreme model sizes improves performance on very small-scale tasks, provided the model is sufficiently pre-trained, contrasts with earlier findings that focused on the quality of data and architecture over sheer model size.

The paper's own evidence supports this claim by demonstrating improved performance on small-scale tasks with sufficiently large models pre

The paper's SQuAD v1.1 result establishes that BERT does not benefit at all from external question-answering data because the authors conducted experiments where they compare BERT's performance with and without the addition of external data, showing no significant gain.

The masking-strategy ablation (Table 8, Section C.2) supports the claim that BERT's robustness to different masking strategies is high, but the caveat is that significant performance degradation occurs when the masking procedure deviates too far from the original 80/10/10 strategy, especially when the ratio of masked to unchanged tokens is not respected.

The logical chain from the introduction to ablation evidence in Section 5 can be summarized as follows: Problem identified in the introduction (limited context understanding in standard language models), motivation to introduce BERT with unidirectional models being sub-optimal for sentence-level tasks, design involving two unsupervised tasks (MLM and NSP) aimed at improving context understanding, method applying these tasks to pre-training and then fine-tuning on specific tasks, experiment demonstrating improved performance on various benchmarks, and evidence showing that the combination of MLM and NSP tasks leads to significant improvements over previous models, especially for tasks requiring deep understanding of context.

If the Next Sentence Prediction task were removed entirely from pre-training, the paper's own evidence suggests that

If the Next Sentence Prediction task were removed entirely from pre-training, the paper's own evidence suggests that performance on downstream tasks like Question Answering and Natural Language Inference would suffer considerably. This is based on the comparison between "LTR & No NSP" and "No NSP" in Table 5, where the former (LTR & No NSP) results in a notable drop in MRPC accuracy and SQuAD F1 score, indicating the critical role of NSP in leveraging pre-training gains.

The logical chain from BERT's introduction to evidence against removing NSP demonstrates that the task is integral to BERT's architecture, enhancing its ability to understand sentence-level relationships, which is crucial for a wide range of downstream tasks.

The comparison between "No NSP" and "LTR & No NSP" highlights the impact of the NSP task on the model's learning ability, showing that BERT's ability to learn sentence-level context and relationships is directly tied to this task.

The implication is that BERT's effectiveness across various tasks is significantly tied to its ability to understand the relationships between sentences, which is facilitated by the NSP task. This understanding is foundational for tasks like Question Answering and Natural Language Inference, where context and sentence-level coherence are crucial.

Removing the NSP task from BERT's pre-training process would likely result in a model that struggles with tasks requiring comprehension of sentence relationships, underscoring the importance of NSP in capturing the full context of language and enabling BERT to perform at high levels on a wide array of NLP tasks.

The paper's evidence for the effectiveness of BERT's architecture, particularly the NSP task, is compelling as it directly ties the removal of NSP to a measurable decline in performance across multiple benchmarks, demonstrating that the model's ability to leverage full context is critical for its success in NLP tasks.

The methodology of systematically removing components like NSP to evaluate their impact provides a clear demonstration of the importance of each part of BERT's architecture, contributing to a deeper understanding of what makes BERT effective.

This investigation into the NSP task not only highlights its importance but also suggests that a modular approach to model design, where each component can be individually evaluated and its impact measured, can lead to more transparent and interpretable models.

In conclusion, the evidence presented in the paper strongly supports the necessity of the NSP task in BERT's architecture for achieving high performance across a range of NLP tasks, reinforcing the importance of understanding sentence-level context and relationships in language understanding.